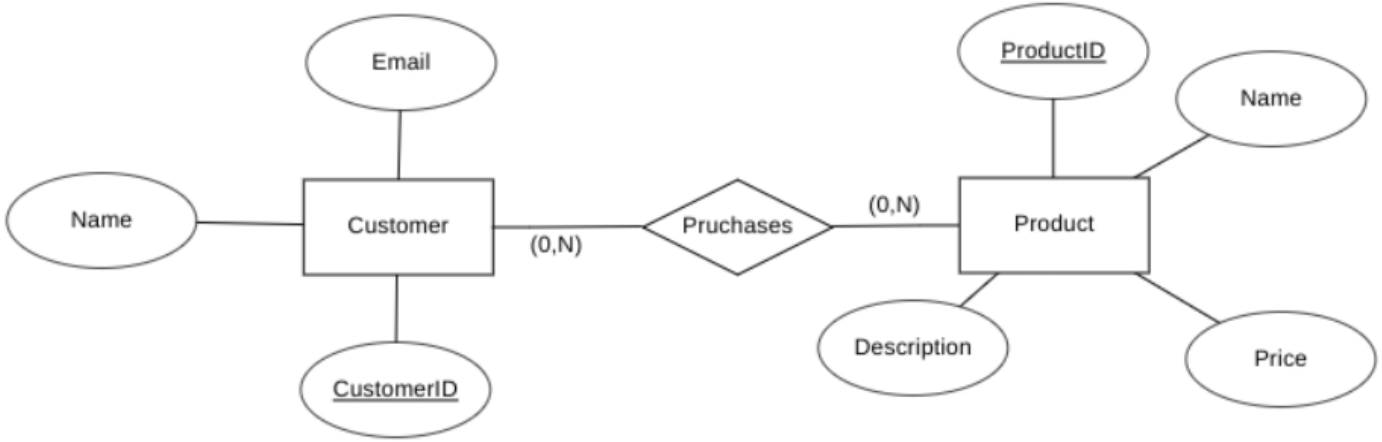
Practice Problems: Basic ER Diagram Creation

Question 1: Basic E-commerce Structure

- **Entities**: Customer, Product
- **Attributes**:
  - Customer: CustomerID, Name, Email
  - Product: ProductID, Name, Price, Description
- **Relationship**: Purchases
- **Instructions**: Create an ER diagram that represents customers and the products they can purchase.

Practice Problems: Basic ER Diagram Creation (Solution)



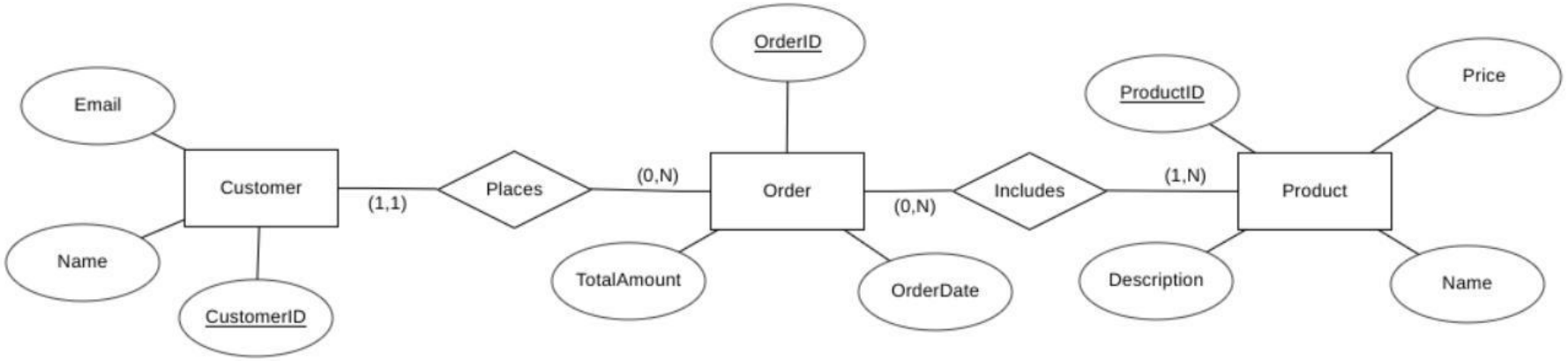
- Notice the Purchases relationship doesn't detail the order process. How can we track order dates, quantities, or multiple products in a single transaction?
- Consider the limitations of representing Purchases without an Order entity. What additional information might be needed for each purchase?

Practice Problems: Introducing Associative Entity

Question 2: Adding Order Processing

- **New Entity**: Order
- **New Attributes**:
  - Order: OrderID, OrderDate, TotalAmount
- **New Relationship**: Places between Customer and Order, Includes between Order and Product
- **Instructions**: Expand on the ER diagram from Question 1 to include orders. Show how customers place orders and how orders include products.

Practice Problems: Introducing Associative Entity (Solution)



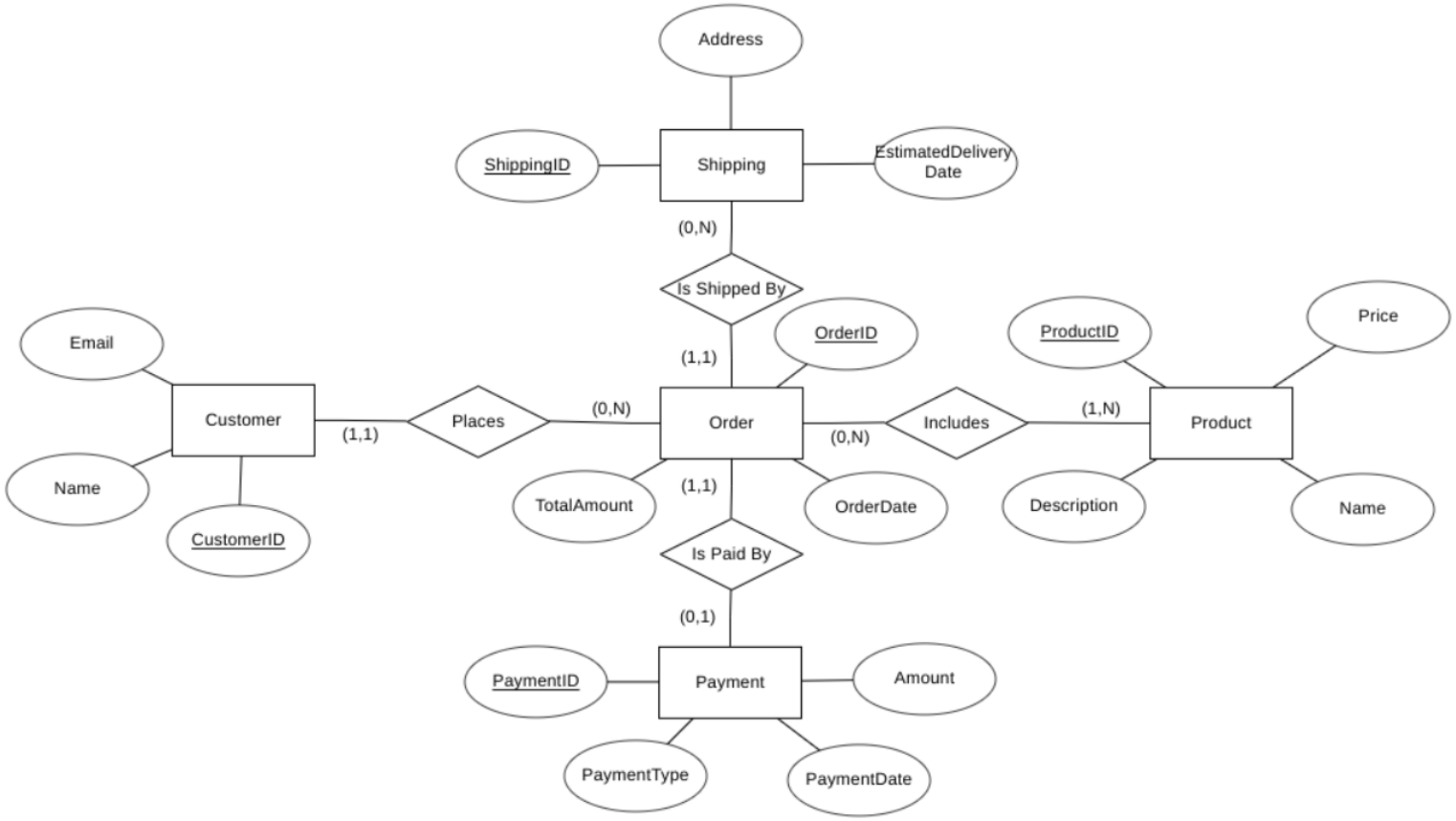
- Currently, the diagram shows orders but lacks details on how payments are processed. How might payments be integrated into the system?
- Consider how orders are fulfilled. What information about shipping or delivery could be added to enhance the model?

Practice Problems: Incorporating Complex Relationships

Question 3: Incorporating Payment and Shipping

- **New Entities**: Payment, Shipping
- **New Attributes**:
  - Payment: PaymentID, PaymentType, PaymentDate, Amount
  - Shipping: ShippingID, Address, EstimatedDeliveryDate
- **New Relationships**: Is Paid By between Order and Payment, Is Shipped By between Order and Shipping
- **Instructions**: Further expand the ER diagram to include payment and shipping details for each order.

Practice Problems: Incorporting Complex Relationships (Solution)



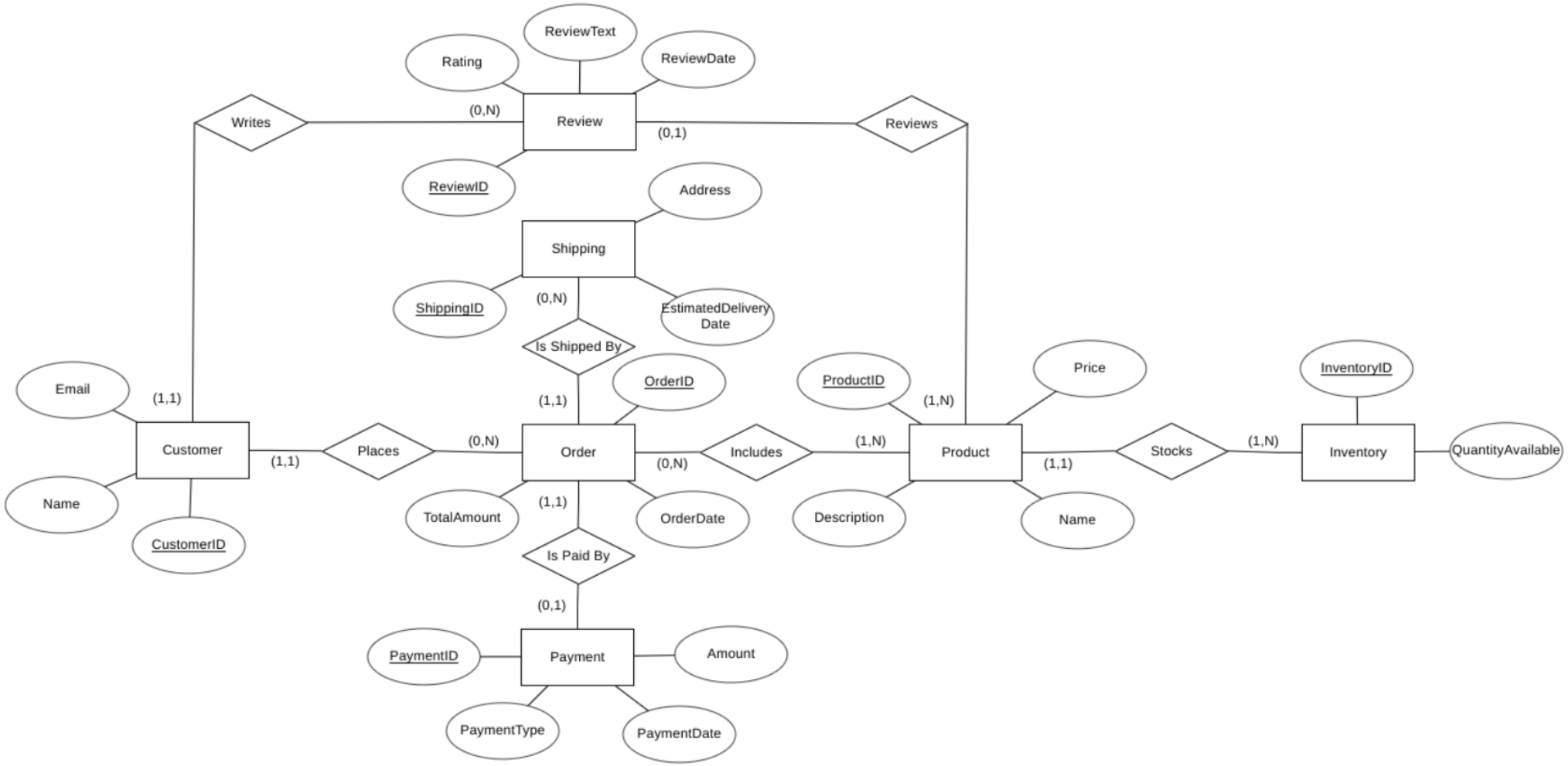
- The diagram now includes payment and shipping, but how can we capture customer feedback on products? Think about integrating product reviews.
- Reflect on inventory management. How does the current diagram track product availability or stock levels?

Practice Problems: Adding Hierarchies and Complex Attributes

Question 4: Advanced Features - Reviews and Inventory

- **New Entities**: Review, Inventory
- **New Attributes**:
  - Review: ReviewID, Rating, ReviewText, ReviewDate
  - Inventory: InventoryID, QuantityAvailable
- **New Relationships**: Writes between Customer and Review (associated with Product), Stocks between Product and Inventory
- **Instructions**: Complete the ER diagram by adding product reviews written by customers and inventory management for products.

Practice Problems: Adding Hierarchies and Complex Attributes (Solution)



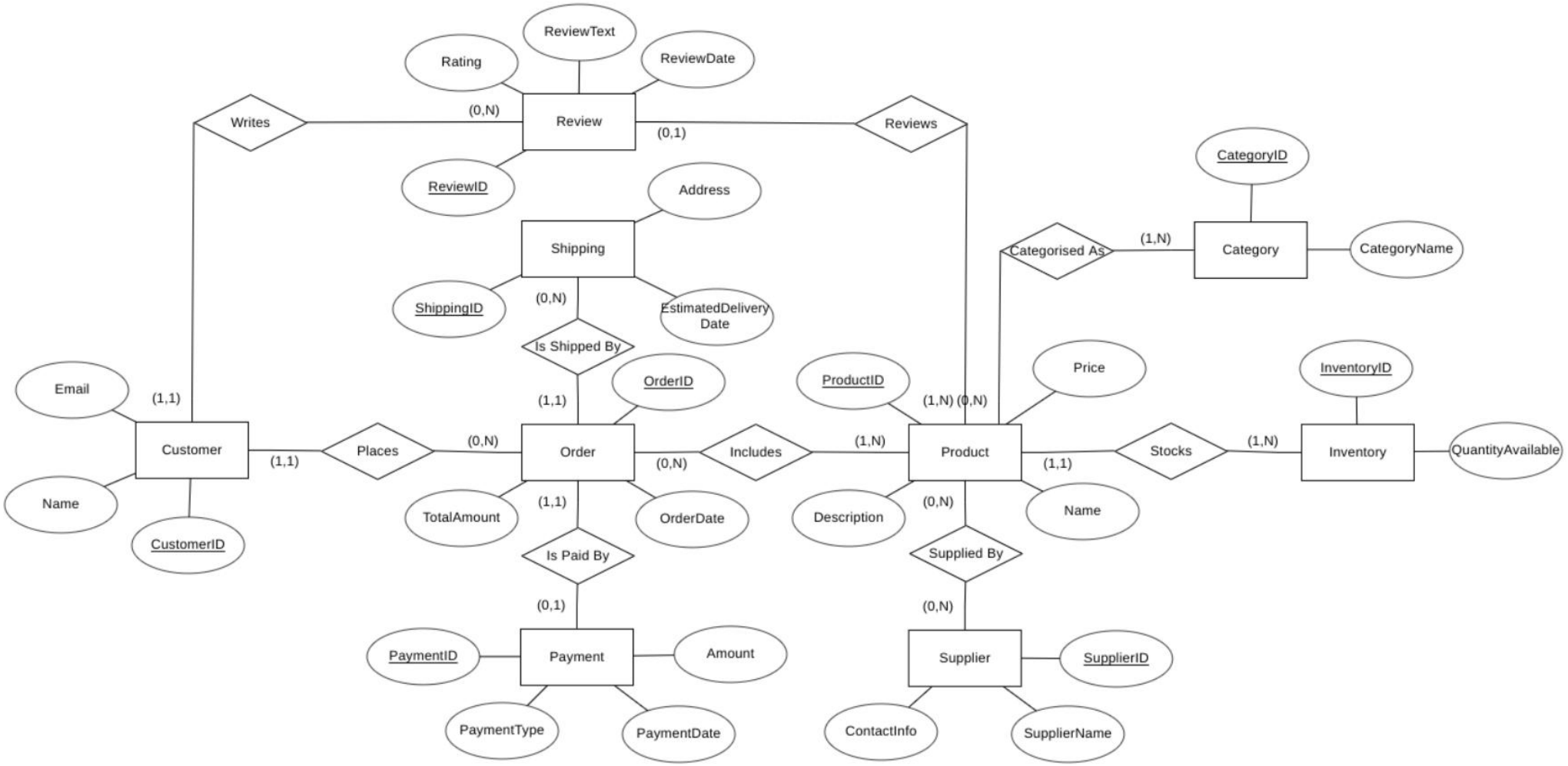
- Our current diagram efficiently handles transactions and inventory, but how are products sourced? Consider the role of suppliers in the e-commerce ecosystem.
- Think about product categorization. How can we represent the various categories a single product might belong to?

Practice Problems: Advanced Hierarchies and Associations

Question 5: Integrating Supplier and Category Management

- **New Entities**: Supplier, Category
- **New Attributes**:
  - Supplier: SupplierID, SupplierName, ContactInfo
  - Category: CategoryID, CategoryName
- **New Relationships**: Supplied By between Product and Supplier, Categorized As between Product and Category
- **Instructions**: Expand the ER diagram to include product suppliers and categories. Show how each product can be supplied by multiple suppliers and belong to multiple categories.

Practice Problems: Advanced Hierarchies and Associations (Solution)



- With suppliers and categories added, consider other real-world aspects. How might we handle returns or exchanges in our e-commerce system?
- Reflect on the user experience. How could the system support personalized product recommendations or wish lists?